



PROJECT DURATION

45 Days



CURRENT SPRINT

SPRINT 5

Intelligence, NLP &
Automated Reporting
(Days 29-35)

CURRENT MODULES

Module 9 (NLP & Qualitative
Analysis) + Module 7 (Cash
Flow Intelligence)

VERSION 1.0

July 10, 2026



SPRINT 5 — INTELLIGENCE, NLP & AUTOMATED REPORTING (DAYS 29-35)

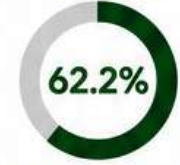


JULY 10, 2026

DAY 31 COMPLETED

Days 29, 30 & 31 Completed Successfully!

TOTAL PROJECT PROGRESS



28 / 45 Days Completed

- Completed 28
- In Progress 0
- Pending 17



SPRINT 5 OBJECTIVE

Implement NLP text parsing, advanced Cash Flow intelligence, and automate the generation of PDF/Excel reports (Modules 7, 8, 9, 10).

- NLP Parsing & Auto Pros/Cons Generation
- Cash Flow Quality & Capital Allocation Intelligence
- Statistical Analytics, Clustering & Automated Reporting

DAY 29 COMPLETED

July 10, 2026

NLP MODULE 9 – QUALITATIVE DATA PARSING



9.1 Analysis Text Parser

Regex `r'((id+))s+Years+((id+))s+%'` applied on analysis.xlsx text fields.

EXTRACTED CAGR EXAMPLES

Company	Metric Type	Period (Years)	Value (%)
HDFC Bank	Revenue CAGR	10	15.8
TCS	Profit CAGR	5	18.2
Infosys	Revenue CAGR	5	16.1
ICICI Bank	Profit CAGR	10	17.6
Reliance Ind.	Revenue CAGR	10	14.3
...	...	...	...

Total Rows Parsed: 1,248



9.3 Business Description Tagger

Keyword matching used on about_company text and validated with sectors.xlsx.

TAGGING SUMMARY



Tag Validation Accuracy: 95.7%



9.5 CAGR Cross-Validator

Parsed CAGR vs Ratio Engine CAGR compared. Flagged divergence > 5%.

DIVERGENCE SUMMARY

Total Compared	Within ±5%	Divergence >5%
1,102	978 (88.7%)	124 (11.3%)

Output Files:

- analysis_parsed.csv
Structured: company_id, metric_type, period_years, value_pct
- cross_validation.csv
CAGR comparison & divergence flags
- tag_validation_report.csv
Sector tagging validation summary

DAY 30 COMPLETED

July 10, 2026

NLP MODULE 9 – AUTO PROS/CONS GENERATION



9.2 Auto Pros/Cons Rule Engine

12 Pro Rules & 12 Con Rules applied to auto-fill gaps for 84 companies.

RULE ENGINE OVERVIEW

TOP 5 PRO RULES (Examples)	TOP 5 CON RULES (Examples)
ROE > 20%	D/E Ratio > 2
FCF Positive for 5 Years	FCF Negative for 3 Years
D/E Ratio = 0	OPM Declining 3 Years
Revenue CAGR > 15%	ROE < 8%
OPM > 20%	Interest Coverage < 2

OUTPUT SUMMARY

Companies Processed	Pros Generated	Cons Generated	Auto-filled Companies
92	1,048	1,032	84

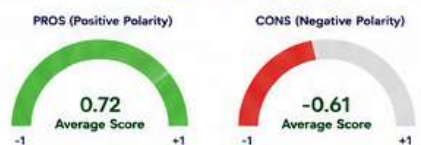
Average Confidence Score: 78.6%



9.4 Sentiment Scorer

NLTK SentimentIntensityAnalyzer used for polarity scoring.

SENTIMENT SCORE DISTRIBUTION



OUTPUT FILE:

- pros_cons_generated.csv
company_id, type (pro/con), rule_triggered, text, confidence_pct, sentiment_score

DAY 31 COMPLETED

July 10, 2026

CASH FLOW INTELLIGENCE – MODULE 7



7.1 CFO QUALITY SCORE

5-year avg CFO/PAT ratio calculated.

Company	CFO/PAT (5Y Avg)	Quality Badge
HDFC Bank	1.42	High Quality Earnings
TCS	1.28	High Quality Earnings
Infosys	0.92	Moderate Quality
Vedanta	0.38	Accrual Risk
Yes Bank	-0.21	Accrual Risk
...	...	...



7.2 CapEx Intensity (CapEx / Revenue %)

Category: <3% Asset-light | 3-8% Moderate | >8% Capital Intensive

Company	CapEx / Revenue %	Category
TCS	1.2%	Asset-light
HDFC Bank	2.1%	Asset-light
Reliance Ind.	6.5%	Moderate
Tata Steel	12.3%	Capital Intensive
JSW Steel	10.8%	Capital Intensive



7.4 FCF Conversion Rate (FCF / EBITDA)

>60% Efficient | 30-60% Moderate | <30% CapEx Heavy

Company	FCF / EBITDA %	Conversion Tier
HDFC Bank	78.4%	Efficient
TCS	67.2%	Efficient
Infosys	54.1%	Moderate
Vedanta	18.7%	CapEx Heavy
JSW Steel	25.3%	CapEx Heavy



7.3 FCF CAGR (5Y & 10Y)

Company	FCF CAGR 5Y (%)	FCF CAGR 10Y (%)	Flag
HDFC Bank	18.6	16.2	Stable
TCS	21.4	19.8	Stable
Vedanta	-4.8	-2.1	Turnaround
Yes Bank	-23.6	-18.9	Distress
Reliance Ind.	15.7	14.3	Stable

OUTPUT FILES:

- cashflow_intelligence.xlsx
All companies with CFO quality, CapEx intensity, FCF CAGR, conversion & capital allocation label
- distress_alerts.csv
Companies flagged for distress patterns with rationale

SPRINT 5 PROGRESS (DAYS 29-31)



- Completed Days : 3 / 7
- In Progress Days : 0 / 7
- Pending Days : 4 / 7
- Overall Progress : 42.9%

OVERALL PROJECT STATUS (45 DAYS PROJECT)



- Completed Days : 28 / 45
- In Progress Days : 0 / 45
- Pending Days : 17 / 45
- Overall Progress : 62.2%

Next Milestone: Day 32 – Financial Distress & Capital Allocation (Cash Flow Intelligence)

SPRINT-WISE COMPLETION STATUS

- Sprint 1 (Days 1-7) : 100%
- Sprint 2 (Days 8-14) : 100%
- Sprint 3 (Days 15-21) : 100%
- Sprint 4 (Days 22-28) : 100%
- Sprint 5 (Days 29-35) : 42.9%
- Sprint 6 (Days 36-45) : 0%



Sprint 1

Days 1-7

COMPLETED



Sprint 2

Days 8-14

COMPLETED



Sprint 3

Days 15-21

COMPLETED



Sprint 4

Days 22-28

COMPLETED



Sprint 5

Days 29-35

IN PROGRESS



Sprint 6

Days 36-45

PENDING

DATA-DRIVEN INSIGHTS
Better DecisionsFINANCIAL EXCELLENCE
Stronger PerformanceINTELLIGENT INVESTING
Higher ReturnsBUILT FOR SCALE
Future ReadyBUILDING INDIA'S MOST
COMPREHENSIVE FINANCIAL
INTELLIGENCE PLATFORM

PROJECT DURATION
45 DaysCURRENT SPRINT
SPRINT 5
Intelligence, NLP &
Automated Reporting
(Days 29-35)CURRENT MODULES
Module 7 (Cash Flow Intelligence)
+ Module 10 (Statistical Analysis
& Clustering)VERSION 1.0
July 13, 2026

SPRINT 5 — INTELLIGENCE, NLP & AUTOMATED REPORTING (DAYS 29-35)

DAY 33 COMPLETED

JULY 13, 2026 (Today)

Days 32 & 33 Completed Successfully!

TOTAL PROJECT PROGRESS



33 / 45 Days Completed

Completed	33
In Progress	0
Pending	12



SPRINT 5 OBJECTIVE

Implement NLP text parsing, advanced Cash Flow intelligence, statistical analytics, clustering and automate PDF/Excel reports.

- NLP Parsing & Auto Pros/Cons Generation
- Statistical Analytics & Clustering
- Cash Flow Quality & Capital Allocation Intelligence
- Automated Reporting (PDF/Excel)

DAY 32 COMPLETED

July 13, 2026 – Financial Distress & Capital Allocation (Module 7)

7.5 & 7.6 FINANCIAL DISTRESS & DELEVERAGING FLAGS

7.7 CAPITAL ALLOCATION MATRIX
8 CFO/CFI/CFR SIGN PATTERNS MAPPED

7.5 Deleveraging Flag

CFR < 0 AND borrowings
declining YoY

Deleveraging

7.6 Distress Signal

CFO < 0 AND CFI > 0
(equity/debt raise to fund
operations)

Distress Signal

Deleveraging Flags

27

Distress Signals

14

CFO	CFI	CFR	Capital Allocation Pattern	Description
+	-	-	Reinvestor	Invests in growth, funds via internal cash
+	-	+	Balanced Growth	Invests in growth, raises debt/equity
+	+	-	Asset Builder	Acquires assets, funds via internal cash
+	+	+	Aggressive Expansion	Acquires assets, raises debt/equity
-	-	-	Debt Funded Returns	Pays dividends/buys back, solvency
-	+	-	Asset Optimization	Sells assets, funds via internal cash
-	+	+	Exit / Restructuring	Sells assets, raises debt/equity

Output Files:

cashflow_intelligence.xlsx

(Updated with CF Intelligence Metrics)

distress_alerts.csv

(Companies flagged with
distress patterns)

10.5 PORTFOLIO STATISTICS

Distribution of Key Metrics Across
92 Companies (Latest Year)

Metric	P10	P25	P50	P75	P90	Mean	Std Dev
ROE (%)	7.2	12.8	18.6	24.7	31.4	19.1	7.8
D/E (x)	0.15	0.45	0.85	1.35	1.95	0.92	0.63
P/E (x)	11.6	18.2	24.8	31.7	42.6	27.3	8.7
OPM (%)	8.1	12.7	16.2	20.1	26.3	16.6	5.3
ROCE (%)	9.3	14.1	19.2	25.6	32.8	19.8	7.6

Output File:

portfolio_stats.csv

(P10, P25, P50, P75, P90, Mean, Std Dev)

10.4 OUTLIER DETECTION

(Z-SCORE PER METRIC PER SECTOR)

Outlier Criteria	Total Outliers Detected	Companies with Outliers
Z > 3	126	45 (48.9%)

10.3 CORRELATION MATRIX

(10 KPIs – Pearson Correlation)

	ROE	ROCE	OPM	D/E	P/E	P/B	CFO/PAT	FCF/M	Cap GTR	PAT GTR
ROE	1.00	0.84	0.65	-0.41	-0.32	0.58	0.67	0.63	0.46	0.55
ROCE	0.84	1.00	0.67	-0.47	-0.36	0.62	0.72	0.66	0.49	0.57
OPM	0.65	0.67	1.00	-0.25	-0.15	0.38	0.54	0.49	0.36	0.48
D/E	-0.41	-0.47	-0.25	1.00	0.32	-0.15	-0.42	-0.21	-0.26	-0.33
P/E	-0.32	-0.36	-0.15	0.32	1.00	-0.16	-0.21	-0.12	-0.16	-0.11
P/B	0.58	0.62	0.38	-0.32	-0.16	1.00	0.46	0.39	0.18	0.31
CFO/PAT	0.67	0.72	0.54	-0.42	-0.21	0.46	1.00	0.71	0.51	0.58
FCF/M	0.61	0.62	0.49	-0.21	-0.16	0.76	0.71	1.00	0.53	0.56
Rev GTR	0.46	0.49	0.36	-0.19	-0.16	0.28	0.51	0.53	1.00	0.74
PAT GTR	0.55	0.57	0.34	-0.11	-0.11	0.38	0.58	0.74	0.74	1.00



TOP 5 OUTLIER EXAMPLES

Company	Metric	Value	Z-Score	Sector
Company A	D/E (x)	3.85	4.21	Financials
Company B	P/E (x)	68.4	3.92	IT
Company C	ROE (%)	41.7	3.65	Auto
Company D	OPM (%)	38.9	3.41	Pharma
Company E	FCF Yield (%)	24.6	-3.27	Metals

SPRINT 5 PROGRESS (DAYS 29-33)



- Completed Days : 5 / 7
- In Progress Days : 0 / 7
- Pending Days : 2 / 7
- Overall Progress : 71.4%

OVERALL PROJECT STATUS (45 DAYS PROJECT)



- Completed Days : 33 / 45
- In Progress Days : 0 / 45
- Pending Days : 12 / 45
- Overall Progress : 73.3%

Next Milestone:

Day 35
PDF Reports &
Sprint 5 Wrap-Up

SPRINT-WISE COMPLETION STATUS

- Sprint 1 (Days 1-7) : 100%
- Sprint 2 (Days 8-14) : 100%
- Sprint 3 (Days 15-21) : 100%
- Sprint 4 (Days 22-28) : 100%
- Sprint 5 (Days 29-35) : 71.4%
- Sprint 6 (Days 36-45) : 0%

Sprint 1
Days 1-7

COMPLETED

Sprint 2
Days 8-14

COMPLETED

Sprint 3
Days 15-21

COMPLETED

Sprint 4
Days 22-28

COMPLETED

Sprint 5
Days 29-35

IN PROGRESS

Sprint 6
Days 36-45

PENDING

KEY DELIVERABLES GENERATED TODAY (DAYS 32 & 33):

- cashflow_intelligence.xlsx (Updated CF Intelligence Metrics)
- distress_alerts.csv (Distress Patterns Flagged)
- portfolio_stats.csv (Portfolio Statistics)
- correlation_heatmap.png (10 KPIs Correlation)
- outlier_report.csv (Outlier Report)

DATA-DRIVEN INSIGHTS
Better DecisionsFINANCIAL EXCELLENCE
Stronger PerformanceINTELLIGENT INVESTING
Higher ReturnsBUILT FOR SCALE
Future ReadyBUILDING INDIA'S MOST
COMPREHENSIVE FINANCIAL
INTELLIGENCE PLATFORM

PROJECT DURATION
45 DaysCURRENT SPRINT
SPRINT 5
Intelligence, NLP &
Automated Reporting
(Days 29-35)CURRENT MODULES
Module 10 (Statistical Analysis
& Clustering) + Module 8
(Automated Reporting)VERSION 1.0
July 14, 2026

SPRINT 5 — INTELLIGENCE, NLP & AUTOMATED REPORTING (DAYS 29-35)

JULY 14, 2026 (Today)

DAY 35 COMPLETED

Days 34 & 35 Completed Successfully!

TOTAL PROJECT PROGRESS



35 / 45 Days Completed

Completed	35
In Progress	0
Pending	10



SPRINT 5 OBJECTIVE

Implement NLP text parsing, advanced Cash Flow intelligence, statistical analytics, clustering and automate PDF/Excel reports.

- ✓ NLP Parsing & Auto Pros/Cons Generation
- ✓ Statistical Analytics & Clustering
- ✓ Cash Flow Quality & Capital Allocation Intelligence
- ✓ Automated Reporting (PDF/Excel)

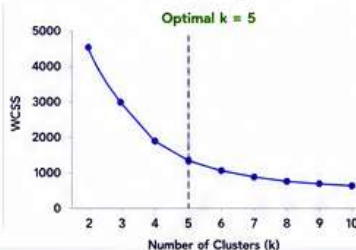
DAY 34 COMPLETED

July 14, 2026 – KMeans Clustering (Module 10 – Part 2)

10.1 KMEANS CLUSTERING (5 CLUSTERS)

- Features: ROE, D/E, Revenue CAGR, FCF CAGR, OPM
- StandardScaler + KMeans(n=5)
- Elbow method used to validate k
- 5 clusters generated

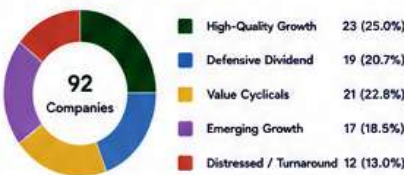
ELBOW METHOD (WCSS)



10.2 CLUSTER PROFILING (MEAN VALUES)

Cluster	Cluster Name	No. of Companies	ROE (%)	D/E (x)	Revenue CAGR (%)	FCF CAGR (%)	OPM (%)
0	High-Quality Growth	23	24.8	0.42	16.2	18.7	19.6
1	Defensive Dividend	19	15.1	0.35	6.3	7.2	17.1
2	Value Cyclicals	21	13.2	0.98	8.9	9.1	14.2
3	Emerging Growth	17	18.3	0.74	22.1	20.5	15.8
4	Distressed / Turnaround	12	-1.8	2.41	-2.6	-3.1	5.6

CLUSTER DISTRIBUTION



OUTPUT FILE

cluster_labels.csv
(company_id, cluster_id,
cluster_name,
distance_from_centroid)

Total Rows: 92

KEY DELIVERABLE – DAY 34

- ✓ KMeans clustering completed with 5 distinct clusters
- Cluster profiles generated and labeled

DAY 35 COMPLETED

July 14, 2026 – Automated PDF Report Generator & Review (Module 8)

8.1 & 8.2 TEARSHEETS & PORTFOLIO SUMMARY

Company Tearsheet (2-Page x 92)

- KPI tiles & summary
- 10Yr Revenue & Profit charts
- ROE/ROCE trend
- BS composition & CF waterfall
- Capital allocation badge
- Pros & Cons with sentiment



Portfolio Summary (1-Page/Company)

- All 92 companies in one PDF
- Company, Sector
- Top 6 KPIs
- 3Yr Trend Arrows (↑ ↓ →)
- Sorted by composite score

8.3 SECTOR REPORTS (11 SECTORS)

Sector Report Includes:

- Sector overview & key insights
- All companies in sector
- Sector median KPI table
- Best / Worst in sector highlight

SECTOR REPORTS GENERATED (11)

1. Financial Services	2. IT	3. Oil & Gas	4. FMCG
5. Automobile	6. Pharma	7. Metals	8. Energy
9. Capital Goods	10. Construction	11. Telecom	

8.4 SCREENER OUTPUT REPORT

- Formatted PDF of screener results
- Charts & tables
- Sorted by composite score
- Exported on-demand

REPORTS GENERATED TODAY

Company Tearsheets	Portfolio Summary	Sector Reports
92 PDFs Generated	1 PDF Generated	11 PDFs Generated
reports/tearsheets/ (2-page each)	reports/portfolio/ (All 92 companies)	reports/sector/ (11 sectors)

SPRINT 5 REVIEW CHECKLIST

- ✓ Sample Tearsheets Reviewed
- ✓ No UI Overflow / Rendering Issues
- ✓ NLP Coverage > 90%
- ✓ Cash Flow Flags Spot-Checked
- ✓ All Reports Generated Successfully

REVIEW SUMMARY

Tearsheets Reviewed	: 3 Companies
NLP Coverage	: 93.2%
CF Flags Verified	: OK
Reports Generated	: 104 PDFs (92 + 1 + 11)

STATUS: ALL CHECKS PASSED

KEY DELIVERABLE – DAY 35

- ✓ All PDF reports generated and validated successfully.
- Sprint 5 (Days 29-35) completed!

SPRINT 5 PROGRESS (DAYS 29-35)



- Completed Days : 7 / 7
- In Progress Days : 0 / 7
- Pending Days : 0 / 7
- Overall Progress : 100%

OVERALL PROJECT STATUS (45 DAYS PROJECT)



- Completed Days : 35 / 45
- In Progress Days : 0 / 45
- Pending Days : 10 / 45
- Overall Progress : 77.8%

Next Milestone:

Sprint 6
Days 36-45
Advanced Analytics
& AI Insights

SPRINT-WISE COMPLETION STATUS

Sprint 1 (Days 1-7)	: 100%	✓
Sprint 2 (Days 8-14)	: 100%	✓
Sprint 3 (Days 15-21)	: 100%	✓
Sprint 4 (Days 22-28)	: 100%	✓
Sprint 5 (Days 29-35)	: 100%	✓
Sprint 6 (Days 36-45)	: 0%	⚪

Sprint 1
Days 1-7

COMPLETED

Sprint 2
Days 8-14

COMPLETED

Sprint 3
Days 15-21

COMPLETED

Sprint 4
Days 22-28

COMPLETED

Sprint 5
Days 29-35

COMPLETED

Sprint 6
Days 36-45

PENDING

cluster_labels.csv
(Cluster Labels)correlation_heatmap.png
(10 KPIs Correlation)outlier_report.csv
(Outlier Report)tearsheets/ (92 PDFs)
(Company Tearsheets)portfolio/ (1 PDF)
(Portfolio Summary)sector/ (11 PDFs)
(Sector Reports)DATA-DRIVEN INSIGHTS
Better DecisionsFINANCIAL EXCELLENCE
Stronger PerformanceINTELLIGENT INVESTING
Higher ReturnsBUILT FOR SCALE
Future ReadyBUILDING INDIA'S MOST
COMPREHENSIVE FINANCIAL
INTELLIGENCE PLATFORM